NOT FOR INTRAVENOUS USE

LAKAN INJECTION

COMPOSITION:
LAKAN INJECTION 1% (10 ML VIAL): Each vial contains Lignocaine Hydrochloride 100 mg per 10 ml
LAKAN INJECTION 2% (10 ML VIAL): Each vial contains Lignocaine Hydrochloride 200 mg per 10 ml

Preservative: Methyl Paraben 0.1% w/v

DESCRIPTION: LAKAN INJECTION is a clear, colourless liquid.

PHARMACODYNAMICS:
Class: Local anaesthetic of the amide type and antiarrhythmic drug. Lignocaine stabilizes all potentially excitable membranes and prevents the initiation and transmission of nerve impulses. This produces a local anaesthetic effect. Onset of action is rapid and blockade may last from 1-1½ hours. In cardiac tissue, a therapeutic serum concentration (1.5 to 6.0 mcg / mL) of lignocaine will produce the following effects.

1. Depression of slow spontaneous depolarization (phase 4), that is the automaticity of isolated, non-polarised Purkinje fibres, while having little effect on conduction velocity, membrane responsiveness or cardiac output. Automaticity induced by stretch, hypoxia or catecholamines can also be suppressed by lignocaine.
2. Shortening of the action potential period and the effective refractory period of Purkinje and ventricular cells. The effective refractory period is not reduced to the same extent, however, and the ratio of effective refractory period to action potential duration is increased.
3. In the presence of high extracellular potassium ion concentrations, or in myocardial ischaemia where the cells are partially depolarized, lignocaine depresses the membrane responsiveness and conduction velocity in Purkinje fibres and the myocardium.

PHARMACOKINETICS:
Absorption: Complete systemic absorption: The rate of absorption is influenced by the site and route of administration, total dosage administered, physical characteristics of the individual agent, and whether or not a vasodilator is used concurrently.
Biotransformation: Lignocaine: Xylidine metabolites are active and toxic, but less so than the parent compound.
Time to peak concentration: Usually 10 to 30 minutes, dependent upon factors affecting rate of absorption. May occur 1 to 3 minutes after intravascular or transtracheal injection.
Peak serum concentration: Depends upon factors influencing the rate of absorption from the injection site and rate of metabolism and distribution volume of the individual anaesthetic.
Elimination: Renal primarily as metabolites: For lignocaine renal excretion may follow biliary excretion into and resorption from the gastrointestinal tract. Quantity of dose excreted unchanged – Lignocaine 10%.

INDICATIONS: For local or regional anaesthesia (not for intravenous use), peripheral nerve block, infiltration anaesthesia and diagnostic or therapeutic block.

RECOMMENDED DOSAGE:
As a local anaesthetic for infiltration and nerve block: The dosage varies and depends upon the area to be anaesthetized, vascularity of the tissues, number of neuronal segments to be blocked, individual tolerance and the technique of anaesthesia. The lowest dose needed to provide effective anaesthesia should be administered. For specific techniques and procedures, refer to standard textbooks.
Adult:

Infiltration:	*Percutaneous:* 5 to 200 mg.
Peripheral nerve block:	*Brachial:* 225 to 300 mg
	Intercostal: 30 mg
	Paracervical: 100 mg per side as a 1% solution; may be repeated if necessary at intervals of not less than 90 minutes.
	Paravertebral: 30 to 50 mg.
Sympathetic nerve block:	*Cervical (stellate ganglion):* 50 mg
	Lumbar: 50 to 100mg

Usual adult prescribing limits: Not to exceed 4.5mg per kg of body weight. In general, the recommended maximum total dose does not exceed 200mg.
Usual pediatric dose: A reduced dosage based on body weight or surface area should be used. The dosage should be calculated for each patient individually and modified in accordance with the physician's experience and knowledge of the patient. Early signs of local anaesthetic toxicity may be difficult to detect in cases where the block is given. In order to minimise the possibility of toxic effects, the use of Lignocaine Injection 1% solution is recommended for most anaesthetic procedures involving paediatric patients. The dose should not exceed 3 mg/kg.

ROUTE OF ADMINISTRATION:
By intramuscular (i.m) or subcutaneous (s.c) route. This solution is not intended for use intravenously. This solution contains preservatives.

CONTRAINDICATIONS: Known hypersensitivity to local anaesthetic of the amide type.

WARNING AND PRECAUTIONS:
Warnings: When lignocaine is used it is mandatory to have emergency resuscitative equipment and drugs immediately available to manage possible adverse effects involving the cardiovascular, respiratory or central nervous system.
Precautions: The safety and effectiveness of lignocaine depend upon proper dosage, correct technique, adequate precautions and readiness for emergencies.

For local anaesthesia:

1. The lowest dosage that results in effective anaesthesia should be used to avoid high plasma levels and serious undesirable systemic side effects.
2. Injection of repeated doses of lignocaine may cause significant increases in blood levels with each repeated dose due to slow accumulation of the drug or its metabolites, or slow metabolic degradation. This is especially relevant in patients with hepatic and / or renal impairment.
3. Tolerance varies with the status of the patient. Debilitated, elderly patients, acutely ill patients and children should be given reduced dosages, commensurate with their age and physical status.
4. Lignocaine should be used cautiously in patients with known drug allergies or sensitivities. Patients showing allergy to ester derivatives of para-aminobenzoic acid such as procaine, benzocaine and tetracaine have not shown cross-sensitivity to agents of the amide type.
5. Lignocaine should be used cautiously in patients with a genetic predisposition to malignant hyperthermia as the safety of anaesthetics of the amide type in these patients has not been established. For epidural anaesthesia, only solutions without preservatives should be used.
6. Local anaesthetics in general should be given cautiously to patients with pre-existing abnormal neurological conditions such as neurological reactions may occur following administration.
7. Careful and constant monitoring of cardiovascular and respiratory (adequacy of ventilation) vital signs and the patient's state of consciousness should be accomplished after each local anesthetic injection. It should be kept in mind at such times that restlessness, anxiety, tinnitus, dizziness, blurred vision, tremors, depression or drowsiness may be early warning signs of central nervous system toxicity.

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INTERACTIONS WITH OTHER MEDICAMENTS:
Anti-arrhythmic drugs: Potentiation of cardiac effects may occur when anti-arrhythmic drugs are used concurrently with lignocaine.
Beta-adrenoceptor drugs: Propranolol and metoprolol reduce the metabolism of lignocaine. It is possible that this effect will be repeated with other beta-adrenergic blockers.
Cimetidine: The clearance of lignocaine is reduced when it is administered in conjunction with cimetidine.
Anticonvulsant drugs: Diphenylhydantoin, phenobarbitone, primidone and carbamazepine appear to enhance the metabolism of lignocaine possibly due to an induction of microsomal enzymes, but the significance is unknown. Diphenyldantoin and lignocaine have additive cardiac depressant effects.
Muscle relaxants: There may be an increased risk of enhanced and prolonged neuromuscular blockade in patients treated concurrently with muscle relaxants (eg: suxamethonium).
Antipsychotics/ 5HT3 antagonists: There may be an increased risk of ventricular arrhythmia in patients treated concurrently with antipsychotics which may prolong the QT interval (eg: pimozide, sertindole, olanzapine, quetiapine, zotepine) or 5HT3 antagonists (eg: tropisetron, dolasetron).
Acetazolamide/ loop and thiazide diuretics: Hypokalemia produced by these agents antagonizes the effect of lignocaine.
Alcohol: Acute, severe alcohol intoxication can centrally depress the cardiovascular system and may thereby prolong lignocaine elimination half-life.

INCOMPATIBILITIES:
Additives may be incompatible although no specific incompatibilities have been reported. The introduction of any additive requires special attention to ensure the incompatibilities do not result. Do not administer unless the solution is clear.

USE IN PREGNANCY AND LACTATION:
Use in pregnancy: The safe use of lignocaine during pregnancy has not been established with respect to adverse effects upon foetal development. Careful consideration should be given before administering lignocaine, particularly during early pregnancy. In obstetrics, if vasopressor drugs are used to correct hypotension the obstetrician should be warned that some oxytoxic drugs have additive hypertensive effects and rupture of a cerebral blood vessel may occur during the post partum period.
Use in lactation: Lignocaine passes into breast milk although it would be unlikely that the amounts passing to the infant would lead to significant accumulation of the parent drug in the infant. The remote possibility of an idiosyncratic or allergic reaction in the breast fed infant from lignocaine remains to be determined.

SIDE EFFECTS/ADVERSE REACTIONS:
Central nervous system: Lightheadedness, drowsiness, dizziness, apprehension, euphoria, tinnitus, blurred or double vision, vomiting, sensations of heat or cold, numbness, twitching, tremors, convulsions, unconsciousness, respiratory depression and arrest.
Cardiovascular system: Hypotension, cardiovascular collapse, and bradycardia which may lead to cardiac arrest.
Neurological system: Adverse neurological reactions occur rarely following the use of local anaesthetics. They may be related to the total dose administered, the particular drug used, the route of administration and the physical status of the patient. Persistent anaesthesia, paraesthesia, weakness, paralysis of the lower extremities and loss of sphincter control may occur.
Allergic reactions: Anaphylactoid reactions, including shock have also been reported rarely.

SYMPTOMS AND TREATMENT FOR OVERDOSE:
Overdosage: In anaesthesia, acute emergencies associated with the use of ligncoaine are normally related to high plasma levels or unintended subarachnoid injection. Toxic symptoms may present following a seemingly normal dose as there is a wide variation in patient response to lignocaine. Treatment of toxicity should be to discontinue administration of lignocaine. Adequate ventilation of the patient (including oxygen if necessary) should be ensured and convulsions should be arrested if present (see Management of Adverse Effects).

Management of Adverse Effects: In the case of severe effects, discontinue use immediately. Institute emergency resuscitative procedures and administer the emergency drugs necessary to manage the situation. For severe convulsions, small increments of diazepam or a short acting barbiturate such as thiopentone should be administered. If not available, pentobarbitone may be given. If convulsions interfere with respiration and are not controlled by specific anticonvulsants, suxamethonium may be given to paralyse the patient, in conjunction with artificial respiration.
Should circulatory depression occur, vasopressor drugs such as ephedrine or metaraminol may be used.

PACKING / PACK SIZE: 10, 25 and 100 vials of 10 ml in a box. (not all pack sizes are available locally)

STORAGE CONDITIONS:
Store below 30ºC. Protect from light.
Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

Lakan Injection 1% (10ml vial) and Lakan Injection 2% (10 ml vial) - *Stability after opening:*
Please store in refrigerator for not more than 4 weeks. Do not freeze. Discard if physical changes observed

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
DUOPHARMA (M) SDN BHD
Lot 2599 Jalan Seruling 59, Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor Darul Ehsan, MALAYSIA

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